# Fresh photographed-curve restart

The historical files remain unavailable. A newly authorized run reconstructs the same exact source curve with fresh secure inputs; no historical measurements are pooled.

| | |
|---|---|
| **Field prime** | 72057980207847001 |
| **Source a, b** | 18456521102804593, 19958176276690439 |
| **Curve order / subgroup order** | 72057979853368804 / 18014494963342201 |
| **Selected rational 2-component** | 46 curves, 90 directed maps, five levels (1/3/6/12/24) |
| **Fresh workload** | 10 generators × 10 shared random scalars; 100 problems per curve |

## Verified reconstruction

PARI/GP 2.17.2 independently counted all 46 curves and their group invariants. Every rational 2-kernel was exhausted. Ninety exact rational curve identities and dual compositions ±[2] were checked symbolically over the actual field. Native C++ checked primality, 4600 transported scalar/order fixtures, 9000 directed-map fixture pairs and 900 homomorphism/dual controls.

The generic cleared-denominator Vélu identity also has evaluated certificate `IDC1hc2aee7cc7e845781` over 2^61−1, with 64 sealed and 32 extra points. It is not a Lean proof or a substitute for the separate exact checks over the actual curve field. The issuer/checker passes the repository's committed worked certificate.

## Preserved scope and limits

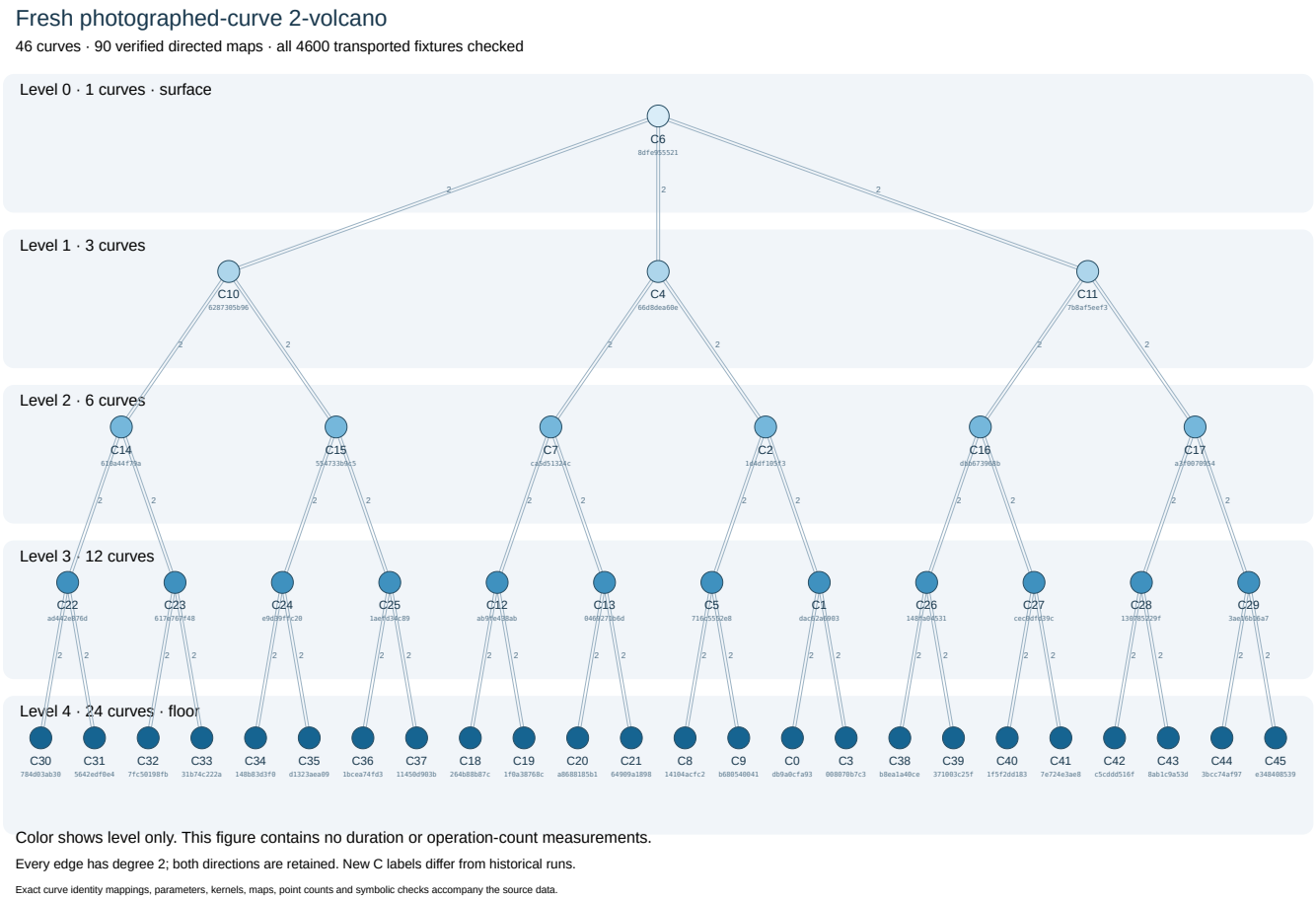
This is the complete selected 2-isogeny component, not the full class for every prime degree. Levels are derived from the closed ordinary volcano structure; the fixed odd conductor is 14549535. C aliases identify this new run only. Full SHA-256 identities and exact parameters are in graph_data.json and SVG titles.

Rho and BSGS execution continues separately toward all 4600 cases per method. Full IC recovery remains unresolved; relation/regularity probes do not count as solved logarithms. No timing, speedup, external-fastest, hardness or full-campaign-completion claim is made in this topology report.

All original fresh records are retained locally. The public archive excludes expected/recovered scalars and raw randomness while publication approval is pending. Private-backup publication is also pending explicit authorization; no complete archive upload is claimed.

**Source:** crypto PR #2056; shortlist commit `ef18058588077ef805c55c5dc25140b6bde2046b`. Native backend: cryptanalysis commit `b7352eec52676b7b82a220ba296661e8ba878915`. Fresh fixture SHA-256: `980d931ce1a804684b160bdab9b213b87688ab9100f0cdf1cd14c390e809acb8`.

# Complete selected 2-volcano




Exact map and identity data accompany this vector diagram. Both directions of every edge are preserved. Level colors are categorical; operation/time heatmaps remain separate measurement views.